

Final Project Report

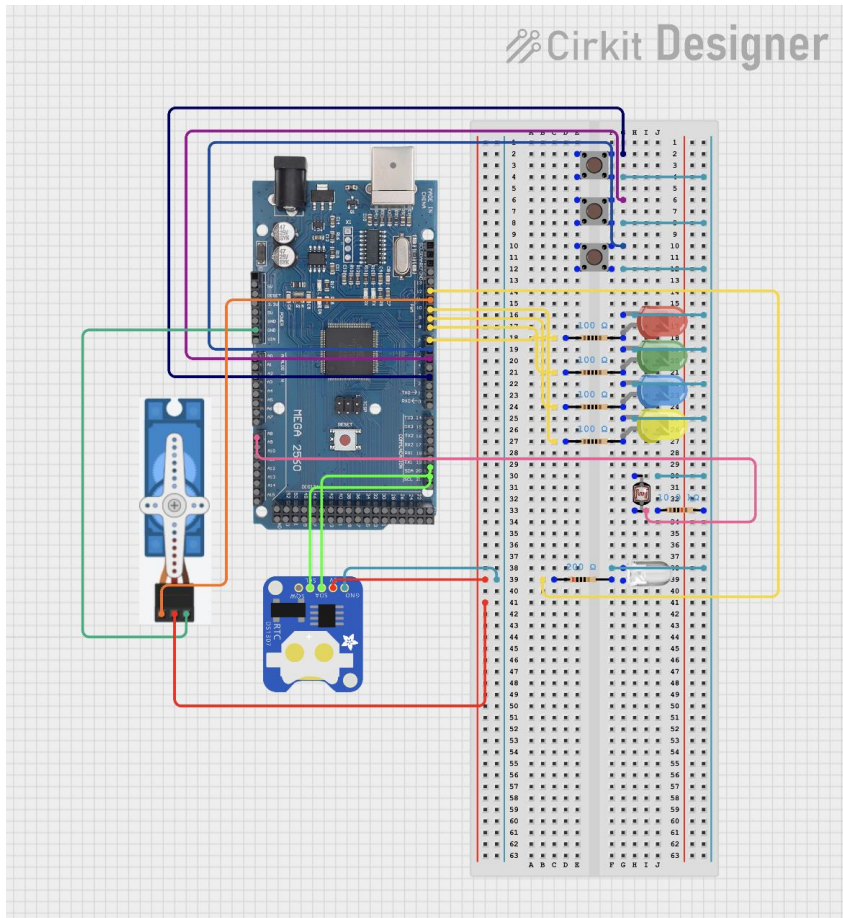
Group 28: Katie Mewes and Maya Muha

Project Motivation/Introduction

The issue we are working to address is the lack of lighting in public and private spaces at night, which causes big safety concerns. The target demographic of this system would be home and business owners wanting to save money on energy costs and create a safer, more well-lit environment. Our system will provide an energy and cost-efficient solution to the lack of lighting at night.

2. Proposed System

2.1 System Schematic Diagram



2.2 Give a detailed component list with references

Arduino Atmega 2560 connected to a computer with the Arduino IDE: This allows the system to communicate back and forth with the computer and supply power to the system.

Solderless Breadboard: This allows for a simple way to organize and connect the systems components.

LEDs(x5): One LED represents our flood light, or a high-powered lightbulb. The other four LEDs are used to represent the different states the system is in. There is a blue for off, a yellow for idle, a green for active and a red for error.

DIP Buttons(x3): There are three push buttons that are wired into the system. The first is the on switch, the second is an off-switch, and the third is a manual reset switch.

Micro Servo 99 SG90: This represents the motor the light resides on that will move the light to the area that needs to be lit.

Photosensor: This collects the light readings of the area and sends the data to the computer.

100 ohm Resistors(x5): These resist the current to the varying parts making sure that they do not overheat/burn out.

10k ohm Resistor: This is used to resist the current to the photosensor.

Jumper Wires: These are used to connect all the components and connect the breadboard to the Arduino.

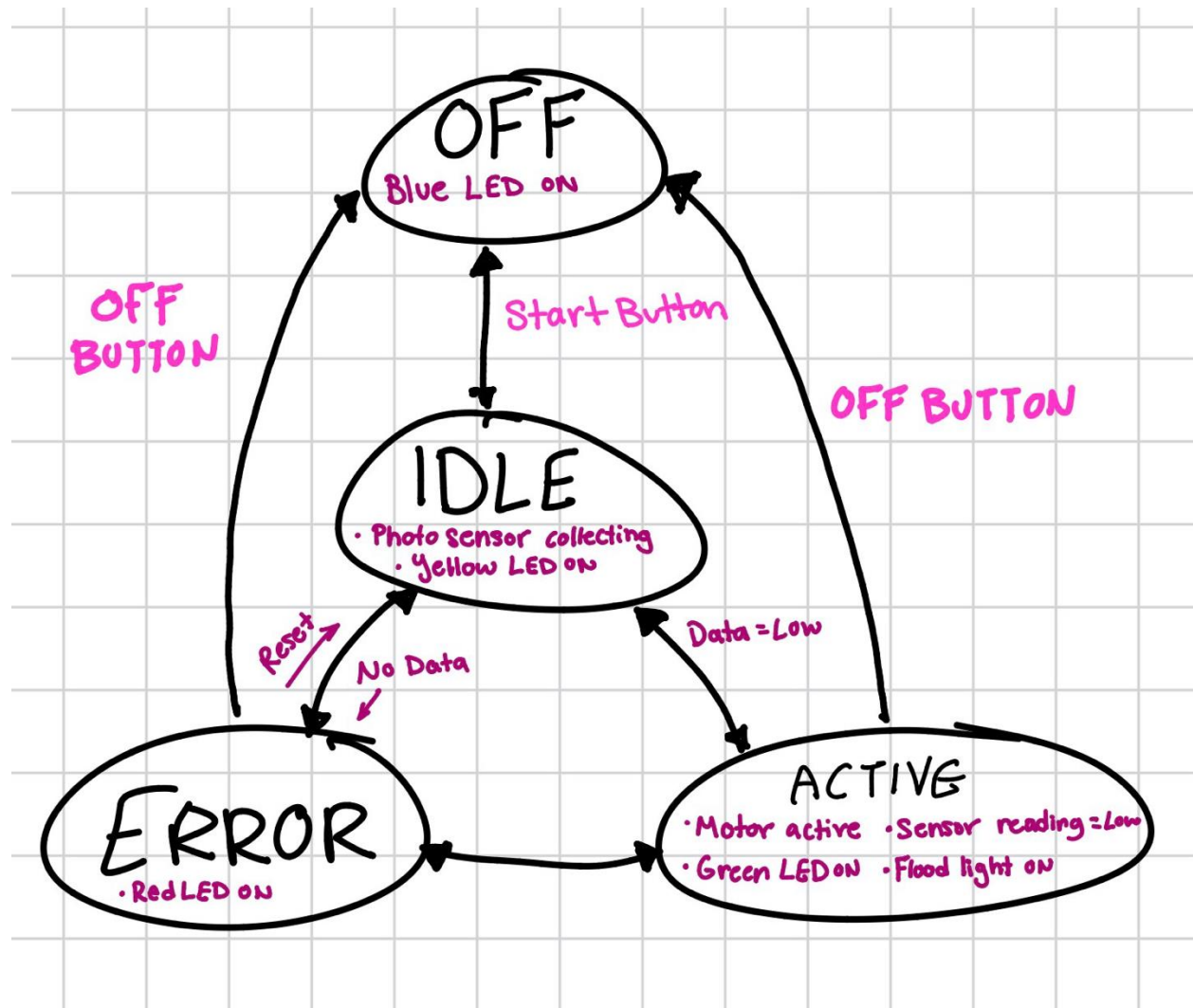
DS1307 RTC Module: Used to track time for serial monitor.

3. System Behavior

3.1 Module Design

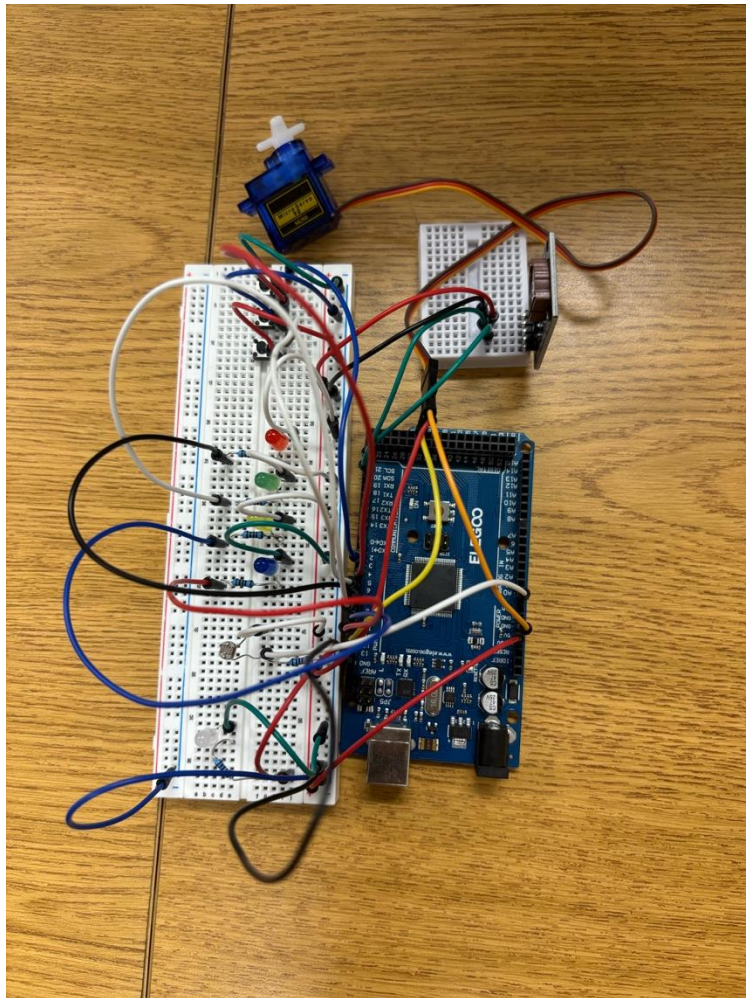
Our system will be divided into multiple modules to organize our code and hardware. A sensor input module will be responsible for detecting light levels. A control logic module will take the information gathered by the sensor input module and decide what state the system should enter, being idle if the light levels detected are high, and turning on if the light levels detected are low. Based on the information from the control logic module, it will get sent to an output module that turns the light on or off. Additionally, we will have a serial communication module responsible for logging important events into the serial monitor.

3.2 Control Logic (State Diagram)

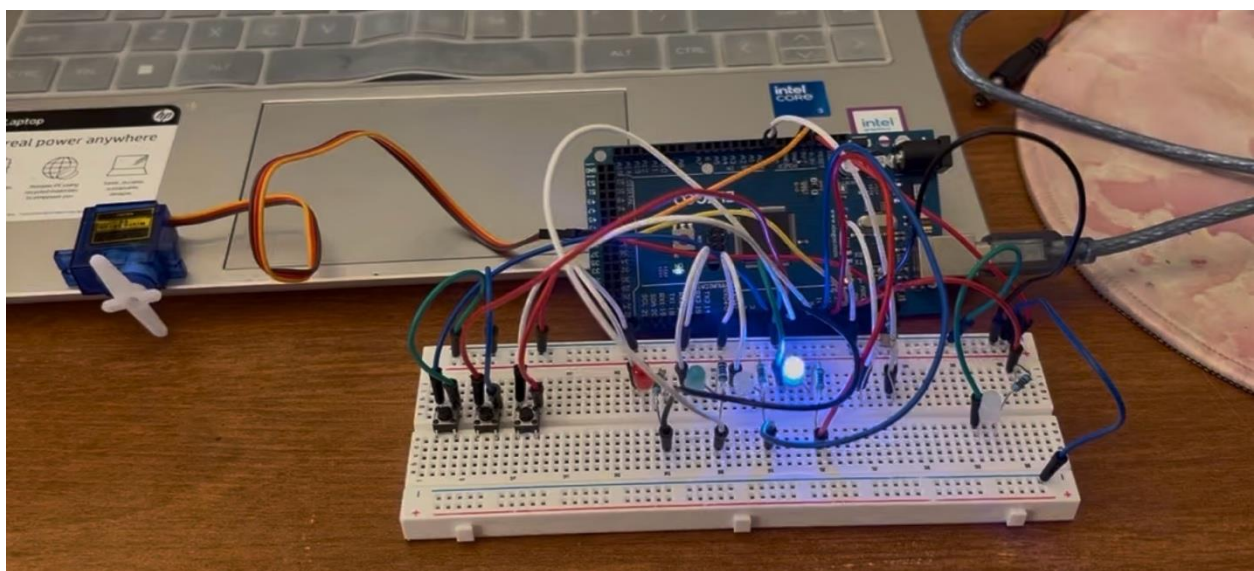


4. Result:

Full circuit image:

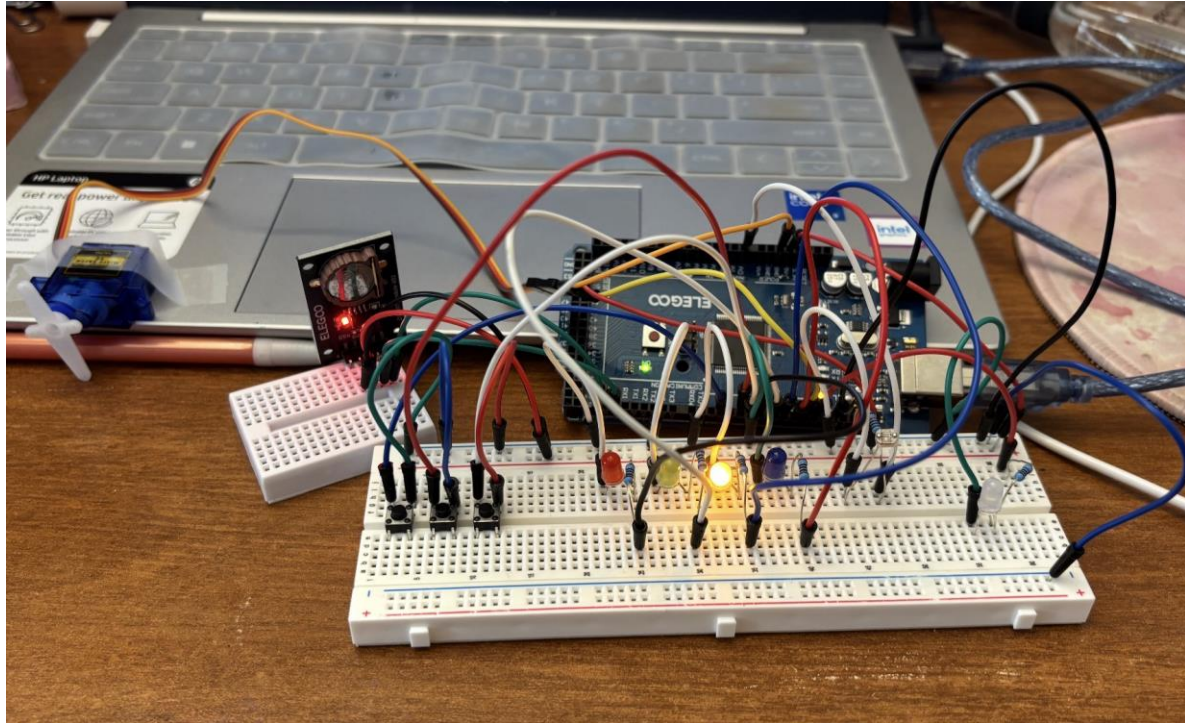


Off State:



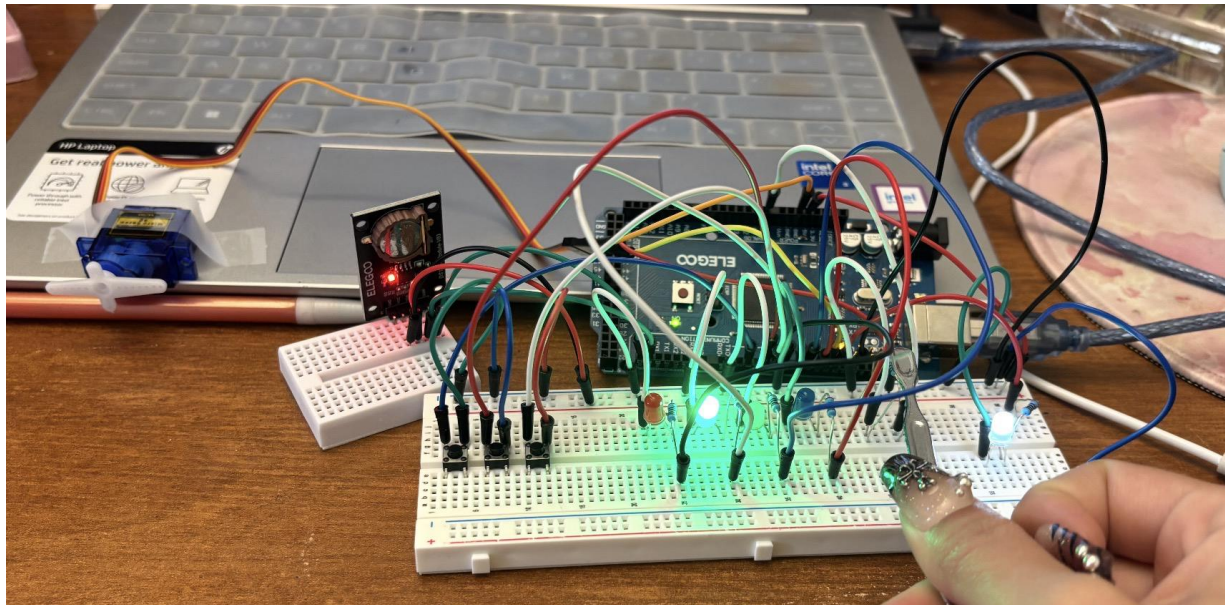
When the Blue LED is on, this signifies that the system is off. No sensor data is currently being collected or analyzed.

Idle State:



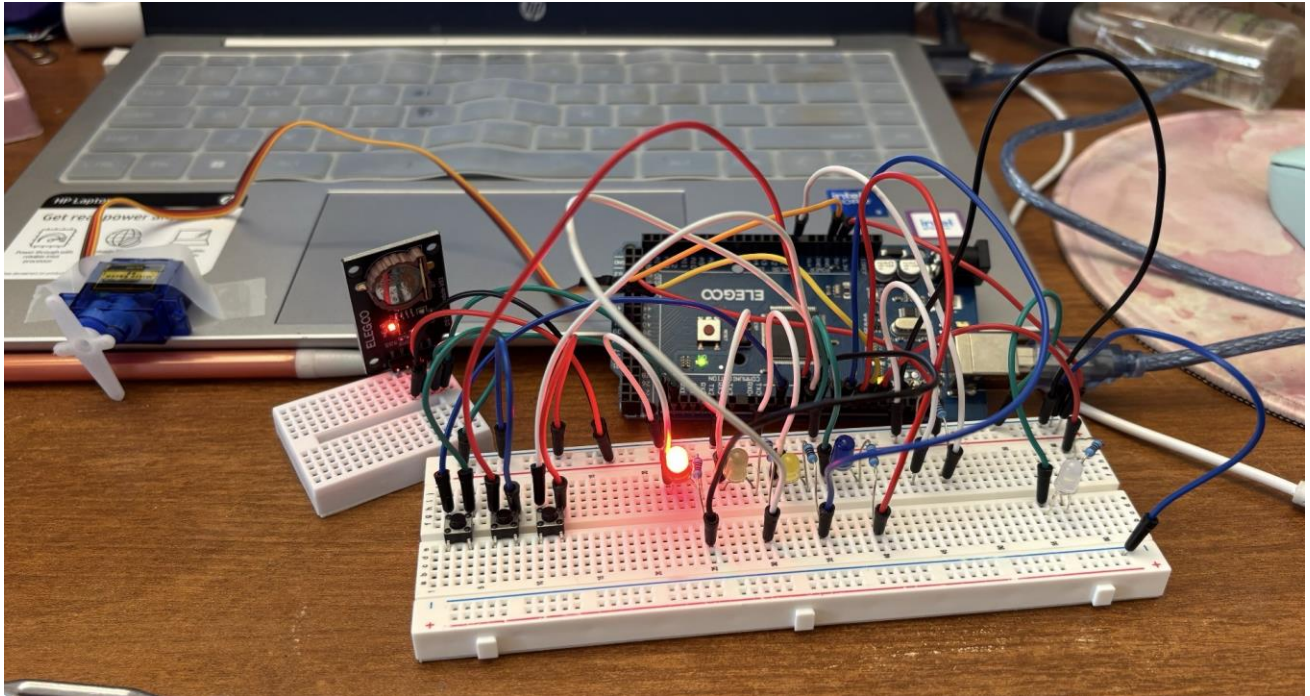
When the Yellow LED is on, this signifies that the system is in an idle state. This means that the sensor is currently receiving and sending data it picks up, but there is enough light to not activate the flood light.

Active State:



When the Green LED is on, this signifies that the system is in an active state. This means that the light reading data picked up from the sensor is below the threshold and that there is light needed to the space. The flood light/white LED is activated.

Error State:



When the Red LED is on, this signifies that the system is in an error state. This means that there is no data being collected by the sensor. This alerts the user to the issue and allows for it to be changed.

Serial Monitor Screenshot:

COM5

18:36:32~IDLE state entered
18:36:44~ACTIVE state entered
18:36:44~Motor 90 position
18:37:3~IDLE state entered
18:37:3~Motor 0 position
18:37:17~ACTIVE state entered
18:37:17~Motor 90 position
18:37:19~ERROR state entered
18:37:41~IDLE state entered
18:37:44~ACTIVE state entered
18:37:44~Motor 90 position
18:37:45~IDLE state entered
18:37:45~Motor 0 position
18:37:47~ACTIVE state entered

☒ Autoscroll ☐ Show timestamp

5. Environmental & Energy Impact

Energy Efficiency: This project minimizes energy waste by only activating when certain conditions have been reached. The light only turns on once it is dark enough outside to justify its usage, thus saving energy by not having it on when it is not needed in daylight.

Design Safety: The system has a manual shutdown button ensuring safety. Additionally if any issues happen with sensor readings, it enters an error state to ensure the people in charge of the light know that the sensor is malfunctioning.

Affordability: This system should be pretty cost efficient to create and distribute because we only need a sensor incorporated and one motor. Since the sensors are simple and don't take in complex input, reducing the cost will not affect the quality in a drastic way.

Sustainability: The system uses cheaper but more efficient products to make sure they will be the most reliable within the budget we have. The system will also be made up of modules, making individual elements easier to repair and replace.

Accessibility: The accessibility for this product is very simple to use. It just needs a power source and then someone to activate the system. Then the system can be deactivated at any time either by cutting power or manually deactivating. This makes the system highly accessible and straightforward to operate.

6. Conclusion

Our system does work efficiently. It is a cheap product to create and takes up little space. The flood light would have to be larger for any real light to be created in a public or private owned area. Some limitations of this project would be weather changes. We created this with a sunny day/ dark night in mind. With any snow/rainfall that could cover the sensor or change the readings, there may be errors. Along with this, the components we used are not weatherproof and will be messed up with natural wear and tear from that kind of weather.